

Yujia Liu - CV

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EDUCATION

Carnegie Mellon University , Pittsburgh, USA Ph.D. in Human Computer Interaction - Advisor: <i>Prof. Alexandra Ion</i>	Aug 2025 – Present
Tsinghua University , Beijing, China M.A. in Information Art and Design - Advisor: <i>Prof. Yingqing Xu and Prof. Chun Yu</i>	Aug 2022 – July 2025
Tsinghua University , Beijing, China BEng in Automation Engineering & BA in Industrial Design (Double Major) - Advisor: <i>Prof. Hong Wang, Prof. Yingqing Xu, and Prof. Lei Zhang</i>	Aug 2017 – July 2022

RESEARCH INTERESTS

Areas	Fabrication, Human-AI Interaction, Education
Methods	Metamaterials, AI Systems, LLMs, Engineering, 3D Printing, Grounded Theory

PUBLICATIONS

- [1] MetaBeads: Enabling Shape- and Stiffness-Changing Interfaces through Beaded Metamaterials.
Yujia Liu, Yuyu Lin, Xie He, Victoria Sun, Trevor Jones, Alexandra Ion
(ACM UIST'26) In *Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology*.
- [2] Towards Accessible Mobility Support: User-Centered Design of a Passive, Multi-Functional, Low-Cost Knee Exoskeleton.
Yuyu Lin, **Yujia Liu**, Emma Kim, Alexandra Ion.
(ACM CHI'26). In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. [\[Link\]](#)
- [3] BrickSmart: Leveraging Generative AI to Support Children's Spatial Language Learning in Family Block Play.
Yujia Liu*, Siyu Zha*, Yuewen Zhang, Yanjin Wang, Yangming Zhang, Qi Xin, Lunyu Nie, Chao Zhang, Yingqing Xu.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [\[Link\]](#)
- [4] Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process.
Siyu Zha*, **Yujia Liu***, Chengbo Zheng, Jiaqi Xu, Fuze Yu, Jiangtao Gong, Yingqing Xu.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [\[Link\]](#)
- [5] Xstrings: 3D printing cable-driven mechanism for actuation, deformation, and manipulation.
Jiaji Li, Shuyue Feng, Maxine Alexandra Perroni-Scharf, **Yujia Liu**, Emily Guan, Guanyun Wang, Stefanie Mueller.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [\[Link\]](#)
- [6] MindShift: Leveraging Large Language Models for Mental-States-Based Problematic Smartphone Use Intervention.
Ruolan Wu, Chun Yu, Xiaole Pan, **Yujia Liu**, Ningning Zhang, Yue Fu, Yuhan Wang, Zhi Zheng, Li Chen, Qiaolei Jiang, Xuhai Xu, Yuanchun Shi.
(ACM CHI'24) In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. [\[Link\]](#)
- [7] KeyFlow: Acoustic Motion Sensing for Cursor Control on Any Keyboard.
Yujia Liu*, Qihang Shan*, Zhihao Yao, Qiuyu Lu.
(ACM UIST'24 Poster) In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. [\[Link\]](#)
- [8] FlexEOP: Flexible Shape-changing Actuator using Embedded Electroosmotic Pumps.
Tianyu Yu, Yang Liu, **Yujia Liu**, Qiuyu Lu, Teng Han, Haipeng Mi.
(ACM UIST'24 Demo) In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. [\[Link\]](#)
- [9] More Than Shapes: Exploring the Tactile Parameters of Art Appreciation for the Visually Impaired.
MingYu Cui, Chao Yuan, **Yujia Liu**, Yingying Zheng.
(ACM UbiComp'24 Workshop) In *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*. [\[Link\]](#)

PROJECT EXPERIENCE

- [P.15] **[Lead] Shape- and Stiffness-Changing Structures Using Beaded MetaMaterials** 10/2025 - 03/2026
 - We developed MetaBeads, a low-cost beading-based fabrication approach for creating shape- and stiffness-changing interfaces using everyday beads and strings.
 - We designed reusable mechanisms, an interactive design tool, and application prototypes across wearables, educational tools, and adaptive furniture.
 - First author of this paper, published at ACM UIST'26.

- [P.14] **[Main Contributor] Passive & Multi-Functional Knee Exoskeleton design** 07/2025 - 09/2025
- We designed a fully passive knee exoskeleton that locks during stance and unlocks during swing using underfoot load as the trigger, enabling more natural walking (including stairs/slopes) without electronics.
 - The prototype combines load-triggered heel/toe switches with a mechanical AND gate and achieves lightweight, low-cost mobility support (0.95 kg, ~\$38) in a sock-like form factor.
 - This paper has been published at ACM CHI'26.
- [P.13] **[Lead] 3D Block Play Instruction Generation and Conducting in real family learning.** 10/2023 - 09/2024
- We designed an AI Agent for children's spatial language learning in family block play, using LLMs and Gen-AI to generate personalized block play instructions for creativity and eco-friendly reuse of bricks.
 - Conducted a lab experiment with 24 parent-child pairs (children aged 6-8), demonstrating the system's effectiveness in enhancing spatial language skills.
 - First author of this paper, published at ACM CHI'25.
- [P.12] **[Lead] LLM Agent for Mentoring Students in the Creative ProblemSolving Process.** 05/2024 - 09/2024
- Co-led the development of Mentigo, an interactive agent using LLMs to mentor middle school students through the creative problem-solving (CPS) process, with dataset of real classroom interactions and an agentic workflow.
 - Tested effectiveness through a comparative experiment with 12 students and feedback from 5 expert teachers, showing significant improvements in student engagement and creative outcomes.
 - Co-first author of this paper, published at ACM CHI'25.
- [P.11] **[Lead] Enhancing AR in Smart Mirrors with Depth-Aligned 3D Visualization** 08/2022 - 09/2024
- Led the development of 3DMirrorcle, a system addressing depth mismatch in AR smart mirrors using lenticular grating for a 3D display and algorithms to align AR content with users' reflections.
 - Conducted user studies across various tasks and scenarios, demonstrating superior performance in user experience compared to existing solutions.
- [P.10] **[Main Contributor] 3D Printing Cable-Driven Mechanism** 06/2024 - 09/2024
- We developed Xstrings, a method for 3D printing cable-driven mechanisms in a single process, enabling four types of interactions: bend, twist, coil, and compress, activated by applying force to the cables.
 - Investigated the impact of various printing parameters on maximum tensile strain and the repeatability of interactions without cable failure.
 - This work has been published at ACM CHI'25.
- [P.9] **[Lead] Using Acoustic Motion Detection for Cursor Control on Keyboard** 05/2024 - 07/2024
- Led the development of KeyFlow, a system that integrates mouse functionality into keyboards using machine learning, enabling users to control the cursor by gliding their fingers across the surface without pressing keys.
 - Our research shows that KeyFlow reduces hand movement by 78.3%, significantly enhancing typing efficiency.
 - This work has been published at ACM UIST'24 Poster.
- [P.8] **[Main Contributor] Flexible Shape-changing Actuator using Electroosmotic Pumps** 04/2024 - 07/2024
- We developed FlexEOP, a method for creating fully flexible electroosmotic pumps, enabling adaptable, self-contained shape-changing actuators. Applications such as flexible displays, panels, curved surfaces, and soft robotic fibers.
 - This work has been published at ACM UIST'24 Demo.
- [P.7] **[Main Contributor] An Aesthetic Education Workshop for the Visually Impaired** 04/2024 - 07/2024
- Experts translated key painting elements (layout, content, color, lighting, brushwork) into tactile forms, and focused on impressionist paintings through workshops, using clay modeling to help participants experience, analyze, and create art, enriching their engagement.
 - This work has been published at ACM UbiComp'24 Workshop.
- [P.6] **[Main Contributor] Leveraging LLMs for Context-Aware Interventions in Digital Wellbeing** 11/2022 - 09/2023
- We developed MindShift, a mobile app that uses LLMs to create dynamic, personalized content aimed at reducing problematic smartphone use, adapting to user context and mental states.
 - Wizard-of-Oz and interview studies were conducted to identify key mental states, and the system was validated in a 5-week field trial with 25 participants, showing significant improvements in smartphone usage patterns.
 - This work was published on ACM CHI'24.
- [P.5] **[Lead] Automated Video Editing with Semantic Analysis and Aesthetic Evaluation** 11/2021 - 04/2023
- We developed an intelligent video editing system that transforms photos and videos from users' smartphones into captivating highlight reels. Using film editing principles, we crafted coherent narratives, emphasized key moments, and ensured seamless harmony between music and visuals.

- [P.4] [Lead] Adaptive Music and Lighting Systems for Emotional Well-being

03/2022 - 03/2023

- We developed a smart home system and music-emotion-light framework that dynamically adjusts music and lighting to enhance the living experience using the Philips Hue system.
- [P.3] [Main Contributor] Design of Tactile Vibration Experience for Smartphones

11/2022 - 02/2023

- We studied vibration experiences across different smartphones and developed a framework that maps task urgency, importance, and metaphorical meaning to vibration timing, duration, intensity, frequency, and variability.
- [P.2] [Main Contributor] User's Color Preferences of Pictures Across Diverse Displays

10/2021 - 12/2022

- This project involved six expert interviews and a user study with 89 participants to identify color preferences for various image types across different smartphone hardware. Then designed a framework to optimize picture color on specific hardware for improved aesthetics and user experience.
- [P.1] [Lead] Ferrofluid Speaker Design Based on Emotion-Mapped Musical Elements [Video]

09/2021 - 06/2022

- Undergraduate graduation project, where I designed and built a ferrofluid speaker that visually responds to music.
 - The ferrofluid inside the speaker moves in sync with the audio, displaying a range of motions, including linear, rotational, and pulsating patterns. These movements dynamically change with the rhythm and sound of the music.

RESEARCH INTERNSHIP EXPERIENCE

Interactive Structures Lab, HCII, Carnegie Mellon University <i>PhD Student / Advisor: Prof. Alexandra Ion</i> Lead the research project of MetaBeads [P.15], and contributed to the research project KneeExo [P.14].	08/2025 - Present
Future Lab, Tsinghua University <i>Research Assistant / Advisor: Prof. Yingqing Xu</i> Led research project: 3D-Mirrorcle [P.11], BrickSmart [P.13], Mentigo [P.12]. Led industry-academic collaboration project: music-lighting [P.4], video editing [P.5]. Contributed to research projects [P.7] and [P.8], industry-academic collaboration project [P.2] and [P.3].	08/2021 - 06/2025
HCI Engineering Group, CSAIL, MIT <i>Visiting Student / Advisor: Prof. Stefanie Mueller</i> Worked on the project of Xstring [P.10], which focuses on 3D printing cable-driven mechanisms in a single process.	06/2024 - 11/2024
Pervasive Interaction Laboratory, Tsinghua University <i>Research Assistant / Advisor: Prof. Yuanchun Shi, Prof. Chun Yu</i> Contributed to the MindShift [P.6], using LLMs to develop interventions for healthier smartphone use.	10/2022 - 06/2023
Tencent, ID/UX Design Group <i>Research Intern / Advisor: Qianhui Liang</i> Engaged in Metaverse project, conducting market research, user analysis, and system design.	07/2021 - 10/2021
Beijing Ewaybot Technology, Robot Navigation Group <i>Summer Intern / Advisor: Bowei Tang</i> Participated in Navigation algorithm research, optimizing code and conducting tests in virtual environments.	06/2020 -08/2020

EXTRACURRICULAR ACTIVITIES

Teaching Assistant, Fundamentals of Computer Literacy	2024
Teaching Assistant, Information Art & Design Trend	2022 - 2023
President, Student Association for Science and Technology, Xinya College, Tsinghua University Led the association to foster academic culture growth through events and innovative promotions.	2017 - 2021
Tsinghua Red Cross Society Engaged in educational support to underprivileged rural children and organized blood donation drives.	2017 - 2018

SKILLS

CS	Python, LLM Implementation, C++, C#, MATLAB, HTML, JavaScript.
EE	Embedded Systems, Arduino, Circuit Design.
Design	Fusion 360, Rhino, Solidworks, Adobe Suite (proficient in PS & PR), Figma, Procreate.
Fabrication	3D Printing, Laser Cutting, CNC, Silicone Casting, Heat Sealing.

REFERENCE

Alexandra Ion (Doctoral advisor) Assistant Professor Director of the Interactive Structures Lab Carnegie Mellon University alexandraion@cmu.edu	Yingqing Xu (Master's advisor) Professor Director of the Future Lab Tsinghua University yqxu@tsinghua.edu.cn	Stefanie Mueller Associate Professor Head of the HCIE Group MIT stefanie.mueller@mit.edu
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